Abstract

Air Handling Units (AHUs) are central components of HVAC systems and account for a significant share of building energy use. Their operation involves several coupled actuators, such as supply and exhaust fans, heating and cooling coils, and dampers. In most buildings, these components are governed by rule-based control strategies that ensure safe operation but often fail to exploit the full flexibility of the system. This master's project aims to develop a reinforcement learning (RL) based control framework for an AHU using a multi-agent approach. Each main actuator will be assigned an individual learning agent, while coordination between agents will ensure consistent system-level performance. The objective is to improve energy efficiency while maintaining thermal comfort, ventilation requirements, and operational constraints. The project will use operational data and/or a validated simulation model of a real AHU. The proposed control strategy will be evaluated against a conventional rule-based baseline to quantify performance improvements.

Expected Student Tasks & Deliverables

The student is expected to take ownership of the following tasks, culminating in a master's thesis and associated deliverables:

1. Comprehensive Literature Review Review existing research on reinforcement learning for HVAC systems, decentralized and multi-agent control approaches, and advanced AHU control strategies.

2. System Analysis and Problem Formulation Study the selected AHU configuration and define control objectives, measurable states, control inputs, and system constraints. Formulate the reinforcement learning control problem.

3. RL Framework Development Design and implement a multi-agent reinforcement learning architecture where each agent controls one actuator. Define appropriate reward functions and coordination mechanisms.

4. Simulation and Evaluation Train and test the proposed control strategy in a simulation environment or using historical operational data. Compare its performance against a standard rule-based controller.

5. Performance Assessment and Documentation Evaluate energy performance, comfort compliance, and control stability. Document the methodology, results, and conclusions in the final thesis.

Required Student Skills and Competencies

The ideal candidate for this project will possess or be highly motivated to acquire the following skills: • Strong background in control systems and dynamic modeling • Proficiency in Python or MATLAB • Basic knowledge of HVAC systems and energy systems • Interest in reinforcement learning and data-driven control

Contact person

Youssef Elomari

Postdoctoral Researcher Department of Civil and Architectural Engineering ABE School,

KTH Royal Institute of Technology yelo@kth.se